

Neuro-Symbolic System-2 Reasoning Architecture for Hallucination Reduction

Project Overview

Project Title

Neuro-Symbolic System-2 Reasoning Architecture for Hallucination Reduction

Project Domain

Artificial Intelligence, Cognitive Science, Large Language Models, Neuro-Symbolic AI

Problem Statement

Large Language Models (LLMs) generate fluent and human-like responses but often fail in multi-step reasoning tasks due to hallucinations and lack of logical validation. Existing approaches primarily rely on prompt engineering or post-generation verification, which do not guarantee reasoning correctness.

This project aims to develop a Neuro-Symbolic System-2 Architecture that combines a Large Language Model with ACT-R (Adaptive Control of Thought – Rational) to enforce structured, step-by-step reasoning and reduce hallucinations before generating the final answer.

Project Goal

Build a reasoning-controlled AI system where:

- LLM acts as System-1 (Intuitive Thinking)
 - ACT-R acts as System-2 (Analytical Thinking)
 - Every reasoning step is validated before progressing
 - Incorrect reasoning is rejected and regenerated
 - Working memory maintains reasoning context
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Core Objectives

1. Integrate ACT-R with an open-source LLM.
 2. Implement working memory buffers for reasoning control.
 3. Create a symbolic validation engine.
 4. Generate structured reasoning outputs.
 5. Reduce hallucination in multi-step reasoning tasks.
 6. Evaluate performance using benchmark datasets.
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Proposed Architecture

Component 1: Large Language Model (System 1)

Purpose:

- Knowledge retrieval
- Natural language understanding
- Candidate reasoning generation

Suggested Models:

- Llama 3
- Mistral 7B
- Phi-3
- Gemma

Input: Structured reasoning prompt

Output: JSON reasoning step

Example:

```
{ "step": 1, "action": "subtract 4 from both sides", "result": "2x = 6" }
```

Component 2: ACT-R Controller (System 2)

Purpose:

- Goal management
- Working memory
- Production rule execution
- Reasoning supervision

Components:

- Goal Buffer
- Imaginal Buffer
- Declarative Memory
- Production Rules

Example Rule:

IF current_goal = solve_equation AND operation_valid = true

THEN proceed_to_next_step

Component 3: Translation Layer

Purpose: Convert symbolic ACT-R states into LLM prompts and convert LLM outputs into symbolic representations.

Flow:

ACT-R State ↓ Prompt Generator ↓ LLM ↓ JSON Output ↓ Parser ↓ ACT-R Validation

Component 4: Validation Engine

Purpose: Verify reasoning correctness before accepting outputs.

Tools:

- Python
- SymPy
- Rule-Based Validator

Validation Checks:

- Mathematical correctness
 - Logical consistency
 - Goal completion
 - Constraint satisfaction
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Working Memory Design

Goal Buffer

Stores:

Current Objective

Example:

Goal: Solve equation

Imaginal Buffer

Stores:

Intermediate reasoning state

Example:

Current Equation: $2x = 6$

Next Action: Divide by 2

Reasoning Workflow

1. User submits problem.
 2. ACT-R stores goal.
 3. Prompt generated for LLM.
 4. LLM produces structured reasoning step.
 5. Validation engine checks correctness.
 6. If valid:
 7. Update buffers
 8. Continue reasoning
 9. If invalid:
 10. Reject output
 11. Regenerate step
 12. Produce final verified answer.
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Unique Features

1. Architecture-level hallucination reduction.
 2. Explicit working memory.
 3. Symbolic reasoning validation.
 4. Controlled reasoning flow.
 5. Regeneration mechanism for incorrect steps.
 6. General-purpose reasoning framework.
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Tech Stack

Frontend:

- React.js
- Tailwind CSS

Backend:

- FastAPI
- Python

AI/ML:

- HuggingFace Transformers
- PyTorch
- ACT-R (pyactr)

Validation:

- SymPy
- Custom Rule Engine

Database:

- PostgreSQL

Version Control:

- Git
- GitHub

Deployment:

- Docker
- Render / AWS

Evaluation Datasets

GSM8K

Mathematical reasoning benchmark

ARC

Abstract reasoning benchmark

Logic Puzzles

Custom evaluation set

Evaluation Metrics

1. Accuracy

2. Hallucination Rate
3. Reasoning Consistency
4. Faithfulness Score

Faithfulness Score:

Verified Steps / Total Steps

Expected Outcomes

- Reduced hallucinations in multi-step reasoning.
 - Improved logical consistency.
 - Transparent reasoning traces.
 - Better faithfulness than standard LLM systems.
 - Reusable neuro-symbolic framework.
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Future Scope

- Medical diagnosis systems
 - Financial decision support
 - Legal reasoning assistants
 - Autonomous agents
 - AI safety and alignment research
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Success Criteria

- At least 25–40% reduction in hallucination rate.
- Higher reasoning consistency than Chain-of-Thought prompting.
- Successful integration of ACT-R and LLM.
- Generation of fully traceable reasoning steps.

This project focuses on building a reasoning-controlled AI system rather than a chatbot, making it suitable for advanced AI research, AI safety, and neuro-symbolic reasoning applications.